Task 5 – Solution Explanation

Summary of the Data Engineering Solution

1. Reading and Cleaning the Data

- **Data Ingestion:** The solution loads the three CSV source files (`customers.csv`, `products.csv`, `orders.csv`) into structured dataframes.
- **Deduplication:** Duplicate customer entries (such as redundant records for customer `C101`) are cleaned using `drop_duplicates()` to ensure data integrity and accurate entity relationships.
- **Orphan Record Handling:** Orders referencing non-existent customers (e.g., order `O105` associated with customer `C105`) are identified and filtered out or isolated to maintain referential integrity.

2. Combining Data and Calculating Metrics

- **Joins:** Valid orders are joined with the cleaned customer dataset (`customer_id`) to fetch customer names and cities, and subsequently with the product dataset (`product_id`) to fetch product names and unit prices.
- **Metric Computation:** The total transaction amount for each order line item is computed using the formula: $$\text{total_amount} = \text{quantity} \times \text{price}$$
- **Final Output Generation:** The transformed dataset is formatted with columns `order_id`, `customer_name`, `city`, `product_name`, `quantity`, `price`, `status`, and `total_amount`, and exported to `final_orders.csv`.

3. Querying & Analysis (SQL)

- Relational table schemas are defined with primary and foreign key constraints.
- Standard SQL aggregations and joins are utilized to answer key business questions, including calculating total customer spending, analyzing order status distributions, identifying top-performing products by sales revenue, and auditing inactive customers.

4. Mapping to AWS Cloud Services

- **Amazon S3** provides scalable storage for raw ingestion and processed outputs.
- **AWS Glue** automates schema discovery (via Crawlers) and executes ETL workloads.
- **AWS Glue Data Catalog** maintains table schemas and metadata.
- **Amazon Athena** enables serverless SQL querying over S3 data for rapid business reporting.
- **AWS IAM** enforces least-privilege security policies across all cloud components.